

level variation. Because this approach nets out *individual*-level targeting, it addresses potential bias from employers directing encouragement toward workers with the highest idiosyncratic benefits. As shown in Table F.4, results remain largely robust under this design.

A.2 Labor Market Outcomes

This section provides a theoretical analysis of how AI chatbots affect labor markets and discusses how these effects can be identified empirically.

Modeling the labor market impacts of AI chatbots raises several key questions: How do chatbots enter the production process? What explains why some firms and workers adopt them while others do not? Are labor markets competitive, or do firms exercise monopsony power? Before committing to a formal framework, Section A.2.1 considers the theoretical implications of each issue.

Section A.2.2 then introduces a model of chatbot adoption and labor market outcomes, embedding the Roy adoption model from Section A.1 into an equilibrium framework in which heterogeneous firms adopt chatbot initiatives and hire workers to maximize profits, while workers sort across firms and adopt chatbots to maximize utility. Section A.2.4 derives predictions for how AI chatbots affect labor market outcomes, and Section A.2.5 applies the model to interpret our difference-in-differences estimates from Section 3.

For clarity of exposition, we focus on encouraged use as the employer initiative, reflecting its central role in the empirical analysis.

A.2.1 Modeling Choices

Chatbot technology. A first key question is how chatbots enter the production process: Are they “co-pilots,” contributing to output only when prompted by humans, or closer to “co-workers,” operating autonomously (Ide and Talamas, 2025)? More broadly, do AI chatbots complement or substitute for labor?

Acemoglu and Restrepo (2018*b*, 2019) provide a framework for how new technologies affect labor productivity through three channels: a negative *task displacement effect*, where

chatbots automate tasks previously performed by humans (e.g., automated copyediting); a positive *task reinstatement effect*, where new tasks for humans emerge (e.g., oversight or integration of AI outputs); and a positive *productivity effect*, where efficiency gains enable firms to scale up production, including labor demand—depending on the elasticity of product demand. Adoption of new technologies increases labor demand only when the combined productivity and reinstatement effects outweigh the displacement effect. A key consideration is whether chatbot use augments or displaces workers’ expertise (Autor and Thompson, 2025); several experiments show that chatbots benefit less experienced workers, helping to close skill gaps within professions (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023).

Modeling chatbots is further complicated by the fact that adoption is not solely a firm-level decision; it also depends on workers’ choices to use the tools. Following Section A.1, workers use chatbots if the productivity benefits exceed the adoption costs. In particular, if the amenity value of chatbot use is sufficiently high, workers may adopt even when doing so reduces their productivity.

In sum, the implications of chatbot adoption for labor demand and productivity are theoretically ambiguous at both the firm and worker levels. At the firm level, the outcome depends on whether the productivity and reinstatement effects together outweigh displacement. At the worker level, it depends on the amenity value of chatbot use.

Firm and worker heterogeneity. Another key question is why some firms and workers adopt chatbots while others do not. Following Mas (2025); Rosen (1986), chatbots can be viewed as a “productive amenity”: firms encourage chatbot use if the profit gains exceed costs, while workers adopt if the productivity benefits outweigh adoption costs (as in Section A.1). In competitive labor markets, Rosen (1986) shows that the wage effects of an amenity depend on the tradeoffs at the margin—the costs and benefits faced by firms and workers just indifferent to providing or accepting it. This implies that the wage effects of chatbot encouragement hinge on the marginal firm and worker.

Labor market competition. Who ultimately benefits from the productivity gains of AI chatbots depends crucially on labor market competition. In competitive markets, workers are compensated only for productivity gains not driven by firm-level investments (Acemoglu and Pischke, 1998; Becker, 1964). Worker-level adoption should then affect wages (Figure 5), whereas firm-level adoption should not. Instead, firm-level adoption should manifest in employment (Figure 8), as firms move down their product demand curves.

An important alternative is the non-discriminatory monopsony model of Card et al. (2018); Robinson (1969), in which firms exert monopsony power because workers have heterogeneous preferences over employers but firms cannot price discriminate across workers. In this framework, any earnings effects from chatbots arise at the firm level through labor demand adjustments, not through within-firm wage differentials between adopters and non-adopters. Thus, within-firm earnings gaps (Figure 5) should not emerge, but firm-level outcomes—total wage bills, employment, and average wages—should be affected (Figure 8). Whether employment effects translate into wages depends on the elasticity of workers’ labor supply across employers—that is, how heterogeneous their firm preferences are.

A.2.2 Formal Setup

This section introduces a formal model of chatbot adoption and labor market outcomes. We make several simplifying assumptions. First, AI chatbots affect output only when actively deployed by workers, reflecting current technology that depends on direct human prompting. Second, workers’ adoption decisions follow the Roy framework of Section A.1, with heterogeneity in both productivity benefits and adoption costs. We apply a parallel Roy model to firms’ encouragement decisions. Finally, product markets are characterized by monopolistic competition and labor markets are perfectly competitive. While the individual components draw on standard frameworks, no existing model combines worker-level adoption decisions with firm-level encouragement policies and equilibrium sorting

across firms. Formalizing these interactions jointly allows us to derive predictions for equilibrium outcomes in Section A.2.4 and to provide structural interpretations of the difference-in-differences estimands in Section A.2.5.

Productivity and earnings. Worker i 's earnings at firm j are the product of a firm-specific skill price W_j and the worker's efficiency units of human capital H_{ij} :

$$Y_{ij}(A_{ij}) = W_j H_{ij}(A_{ij}), \quad (10)$$

where human capital may be augmented through chatbot use, $A_{ij} \in \{0, 1\}$:

$$H_{ij}(A_{ij}) = \exp(B_{ij} A_{ij}), \quad (11)$$

and B_{ij} captures the productivity benefit of AI chatbots. As in Section A.1, productivity benefits and adoption costs depend on worker type and firm encouragement:

$$B_{ij} = \alpha_{b0} + \alpha_{b1} E_j + \beta_b \times i \quad (12)$$

$$C_{ij} = \alpha_{c0} + \alpha_{c1} E_j + \beta_c \times i. \quad (13)$$

Firm production and demand. Firms produce using efficiency units of labor:

$$Y_j = H_j = \int_{i \in \mathcal{I}(j)} H_{ij} di, \quad (14)$$

where $\mathcal{I}(j)$ denotes firm j 's workforce. This formulation models chatbots as tools that augment worker output; Section A.2.1 discusses implications of more general technologies.

Product markets feature monopolistic competition among atomistic, horizontally differentiated firms facing demand curves:

$$Y_j = P_j^{-\epsilon}, \quad (15)$$

where ϵ is the within-sector elasticity of demand.